A

Mini Project Report

on

SeniorConnect

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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CERTIFICATE

This to certify that the Mini Project report on SeniorConnect has been submitted by **Veddika Shetty** (24107131), **Nidhi Tapulli** (24107085) **Smita Shinde** (24107124) and **Mayuri Wagh** (24107063) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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Chapter 1

Introduction

Senior Connect is a smart digital platform designed to support elderly individuals by enhancing their safety, health monitoring, and communication with caregivers. With the increasing aging population and busy lifestyles of family members, seniors often face challenges such as loneliness, delayed medical assistance, and lack of real-time monitoring. This project aims to bridge that gap using modern technologies like mobile applications, emergency alert systems, and health tracking features.

1.1 Purpose

- To provide a user-friendly application for elderly care and assistance
- To enable quick communication between seniors and their family members/caregivers
- To ensure safety through features like SOS alerts and emergency contacts
- To promote independent living while maintaining constant supervision

1.2 Problem Statement

Elderly individuals living independently face higher risks during emergencies due to reduced mobility and lack of immediate assistance. Even minor incidents can become serious if help is delayed.

The absence of **real-time health monitoring** makes it difficult to detect sudden changes in conditions like diabetes or heart issues. Additionally, without **automated alert systems**, emergency response is often delayed as seniors may not be able to seek help themselves.

There is also a **communication gap** with family members, who lack real-time updates on their well-being. Hence, a system with **continuous monitoring and instant alerts** is essential to ensure their safety.

1.3 Objectives

1. **Accessible Daily Check-In Interface**

   Build a Flutter UI with semantic labels, implement a one-tap check-in action, and store status using local persistence with instant state feedback and integrate public health data APIs to detect seasonal flu trends and notify users and caregivers.

2. **Emergency SOS Alert System**

   Implement an SOS module using GPS APIs, (call/SMS/voice note), and audio capture to send real-time location and alerts to predefined emergency contacts via background services.

3. **Medication Reminder and Confirmation Support**

   Set up scheduled notifications with acknowledgment callbacks and securely store medication logs and health notes in a local database with optional cloud sync.

4. **To Integrate Voice Assistance and Accessibility Features**

   Support voice commands, text-to-speech reminders, and adjustable font sizes to enhance accessibility for seniors.

1.4 Scope

- Focus on elderly users with basic smartphone knowledge
- Covers emergency assistance, health tracking, and communication
- Can be extended with text to speech assistants
- Future scope includes integration with wearable devices and smart homes

Chapter 2
Proposed System

The proposed system, Senior Connect, is a mobile-based application designed to provide comprehensive support for elderly individuals through real-time monitoring, emergency assistance, and seamless communication. The system integrates modern technologies such as GPS tracking, health sensors, and cloud-based data storage to ensure safety and reliability. It enables caregivers to stay connected with seniors and receive instant alerts during emergencies.

The application is designed with a simple and intuitive interface to ensure ease of use for elderly users. It also focuses on proactive care by providing health insights and timely notifications. The system ensures quick response in critical situations, thereby improving the overall quality of life for senior citizens.

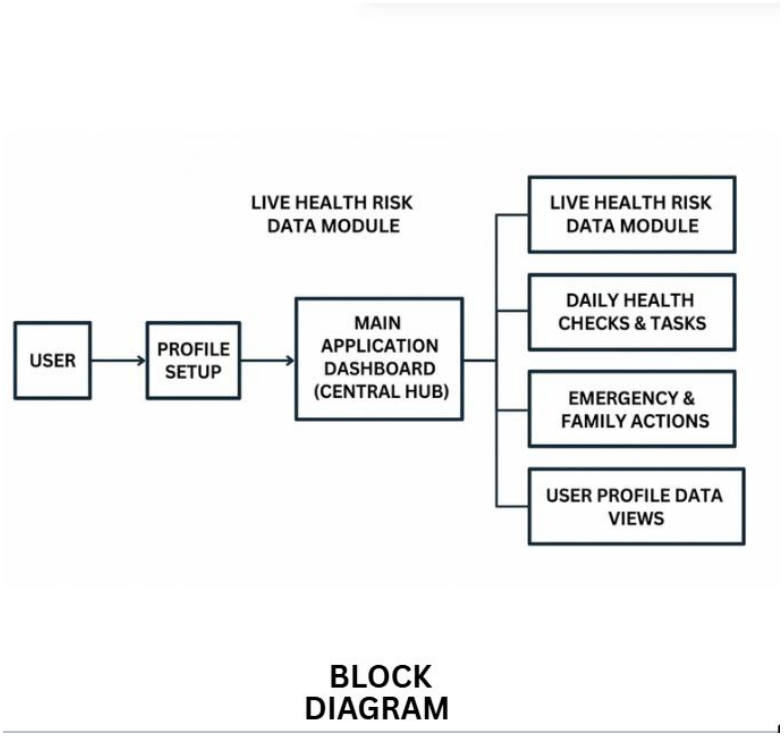

Figure 2.1 Block Diagram

In fig 2.1 the block diagram represents the overall architecture and flow of the SeniorCare system. It begins with the **User**, who interacts with the application by first completing the **Profile Setup**. In this step, essential personal, medical, and emergency contact details are entered, forming the base data required for the system to function effectively.

After the profile setup, the user is directed to the **Main Application Dashboard (Central Hub)**, which acts as the core of the system. This dashboard integrates all functionalities and provides a centralized interface through which the user can access different features of the application.
From the central hub, the system is divided into multiple functional modules. The **Live Health Risk Data Module** fetches real-time information (such as infection levels) to keep users aware of external health risks. The **Daily Health Checks & Tasks** module manages regular activities like check-ins, medication reminders, and routine monitoring.

The **Emergency & Family Actions** module includes features such as SOS alerts and direct communication with family members for quick assistance during critical situations. Lastly, the **User Profile Data Views** module allows access to stored user information, including health conditions and activity history.

Overall, the block diagram illustrates how the system is structured in a modular way, ensuring smooth data flow, centralized control, and efficient functioning of all features to enhance elderly care and safety.

2.1 Features and Functionality

- One-Tap Daily Safety Check-In-Instantly confirm daily safety status with a single tap and stored history.
- Real-Time Emergency SOS System-Send immediate SOS alerts with live location, calls, SMS, and voice notes to emergency contacts.
- Smart Medication Reminder & Tracking-Receive scheduled reminders and log medication intake with simple confirmations.
- Daily Health Notes & Secure Logs- Record and securely store daily health updates for monitoring and reference.
- Text to Voice Assistance -Use voice commands and spoken alerts for hands-free interaction and accessibility.
- Seasonal Flu & Outbreak Alerts-Get notified about flu trends and health outbreaks using public health data sources.

Chapter 3
Project Outcomes

The Senior Connect project successfully resulted in the development of a functional mobile application aimed at enhancing the safety and well-being of elderly individuals. One of the key achievements of the system is the implementation of an SOS emergency alert feature, which allows users to quickly notify caregivers while sharing their real-time location using GPS technology. This ensures immediate assistance during critical situations.

The application also enables seamless real-time communication between seniors and their caregivers, reducing the communication gap and improving response time. In addition, basic health monitoring features such as step tracking and alert notifications have been integrated to provide continuous health insights.

A major focus of the project was on usability; hence, a simple and user- friendly interface was designed specifically for senior citizens, ensuring ease of navigation and accessibility. The system also ensures secure data handling through cloud-based storage, maintaining user privacy and reliability.

Overall, the project demonstrates the effective use of modern technologies such as mobile applications, GPS tracking, and APIs to create a practical and impactful solution that promotes independence, safety, and better quality of life for elderly users.

Chapter 4
Software Requirements

Operating System

- Android OS (for mobile deployment)
- Windows/macOS (development environment)

Development Tools

- Flutter SDK
- Visual Studio Code / Android Studio

Programming Language

- Dart (primary language for app development)

Libraries & Packages Used

- **firebase_core, firebase_auth, cloud_firestore**
  → Used for backend services, authentication, and cloud database
- **geolocator**
  → Used for real-time GPS location tracking
- **flutter_local_notifications**
  → Used for sending alerts and reminders
- **shared_preferences**
  → Used for storing user data locally
- **http**
  → Used for API calls and external communication
- **speech_to_text & flutter_tts**
  → Used for voice interaction (accessibility for elderly users)
- **url_launcher**
  → Used for calling emergency contacts

Chapter 5
Project Design

The workflow of the SeniorCare application begins when the user opens the app. This is the entry point where the elderly individual or user accesses the system through a mobile device. The interface is designed to be simple and user-friendly so that even non-technical users can navigate easily.

After opening the app, the user proceeds to create their profile. In this step, important personal and medical details are entered, such as basic information, health conditions, and emergency contacts. This data is essential as it forms the foundation for monitoring, alerts, and communication features within the system.

Once the profile is created, the user is directed to the Dashboard/Home screen, which acts as the central control panel of the application. From here, multiple functionalities are accessible. The system integrates a Live API to fetch real-time infection levels based on the user's location, helping them stay informed about environmental health risks. The Daily Check-in feature allows users to regularly confirm their safety status, which can be shared with family members.

The dashboard also includes an SOS Emergency Button, which enables users to quickly send alerts and request help during critical situations.

Data Flow

User Action → Flutter App → Firebase Backend → Notification Trigger → Caregiver

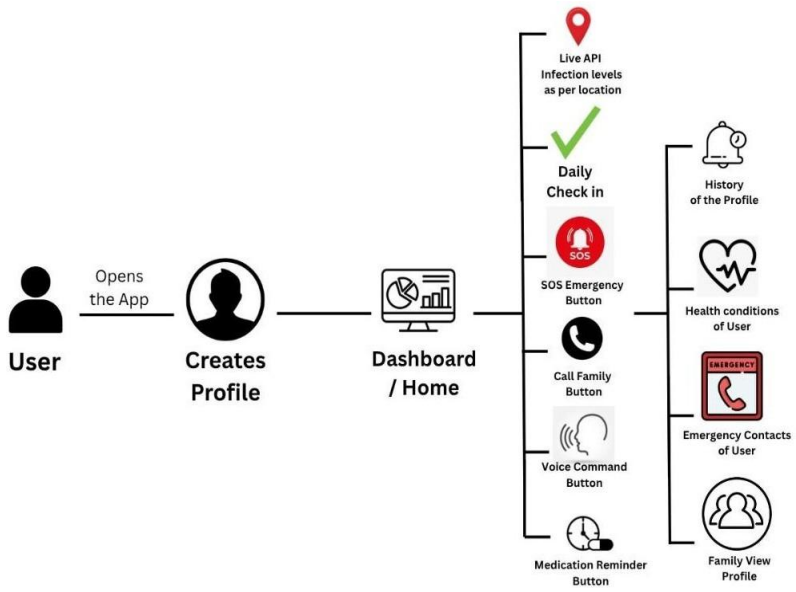

Figure 5.1 Work Flow Diagram

Additionally, the application provides a Medication Reminder Button, which ensures that users take their medicines on time by sending timely alerts and confirmations. This helps in maintaining proper health routines, especially for users with chronic conditions.

On the right side of the workflow, in fig 5.1 the system maintains and utilizes stored data such as History of the Profile, which keeps track of user activities and logs. The Health Conditions of the User section stores medical details for quick reference during emergencies. The Emergency Contacts list is used when SOS alerts are triggered, ensuring immediate communication. Finally, the Family View Profile allows family members to monitor the user's status, improving transparency and caregiving.

Overall, this workflow ensures a smooth flow from user entry to real-time monitoring, emergency handling, and family connectivity, making the system efficient, reliable, and suitable for elderly care.

Chapter 6
Project Scheduling

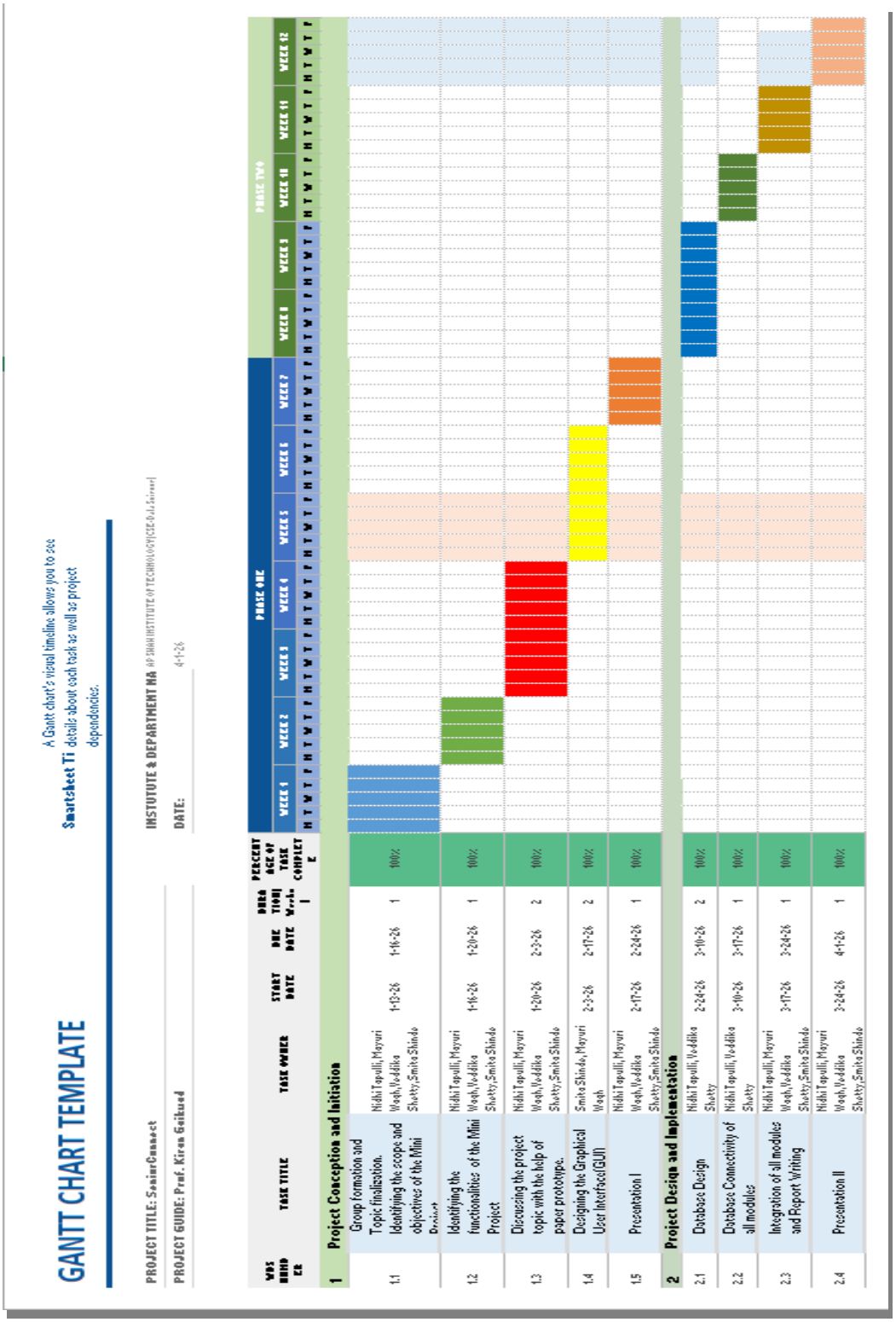

Fig 6.1 Gantt Chart

In fig 6.1 the given Gantt chart represents the week-wise planning and execution of our project. In the initial weeks (Week 1–Week 3), we focused on project conception and initiation, where we finalized the project topic, performed requirement analysis, and defined the overall scope of the system. During Week 4–Week 6, we worked on understanding system functionalities and began designing the basic structure of the project.

From Week 7–Week 10, we moved into the design phase, where we developed system architecture, designed modules, and planned the user interface. Between Week 11–Week 14, the focus shifted to implementation, including database design, backend development, and establishing connectivity between different modules.

In Week 15–Week 17, we carried out integration of all modules and ensured that the system worked as a complete unit. Following this, Week 18–Week 20 were dedicated to testing and debugging, where we identified and fixed errors to improve system performance.

Chapter 7
Results

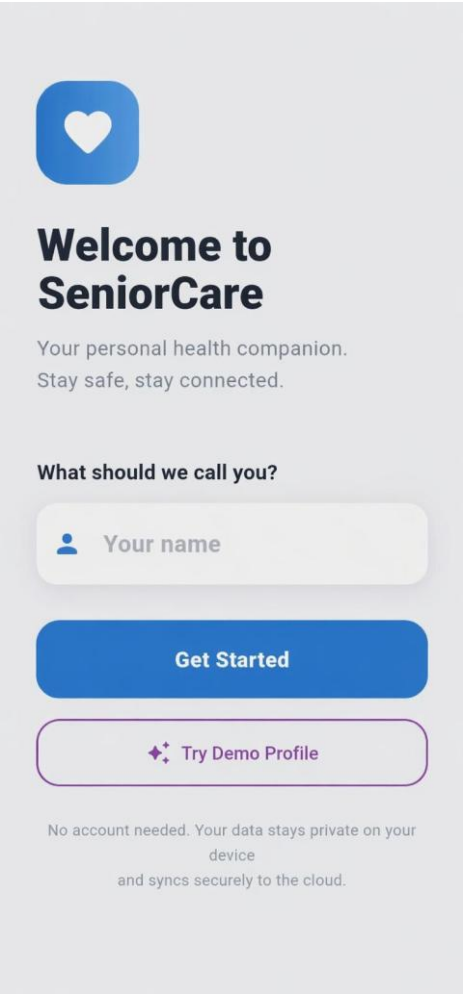

Figure 7.1 Login Page Of SeniorCare

In fig 7.1 the welcome screen acts as the initial entry point of the SeniorCare application and is designed with a simple, clean, and user-friendly interface suitable for elderly users. It displays a welcoming message along with a tagline that emphasizes safety and connectivity, clearly conveying the purpose of the app as a personal health companion. The screen asks the user, “What should we call you?” and provides an input field to enter their name, enabling personalization of the user experience. A prominent “Get Started” button allows users to proceed with the application setup. Additionally, a “Try Demo Profile” option is available for users who wish to explore the app without entering personal details. The screen also reassures users about data privacy and security, helping build trust and confidence in using the application.

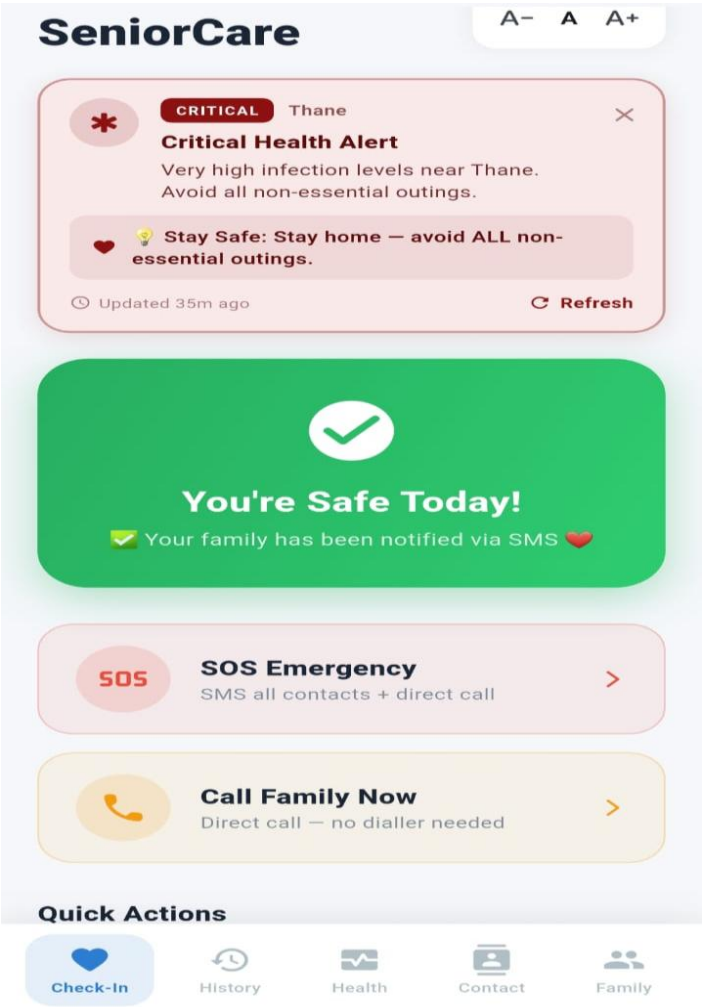

Figure 7.2 SeniorCare Home Dashboard

In fig 7.2 the top section of the SeniorCare application dashboard displays the overall safety status and critical alerts for the user. It includes a Critical Health Alert with location information (such as Thane), indicating high infection levels in the area and advising users to avoid non-essential outings. This alert is visually highlighted to draw immediate attention and ensure the user is aware of potential risks.

Below the alert, the application shows a confirmation message stating “You’re Safe Today!”, which reassures the user about their current condition. Along with this, the system indicates that family members have been notified via SMS, ensuring that the user’s safety status is communicated to close contacts. This feature enhances trust and provides an added layer of security.

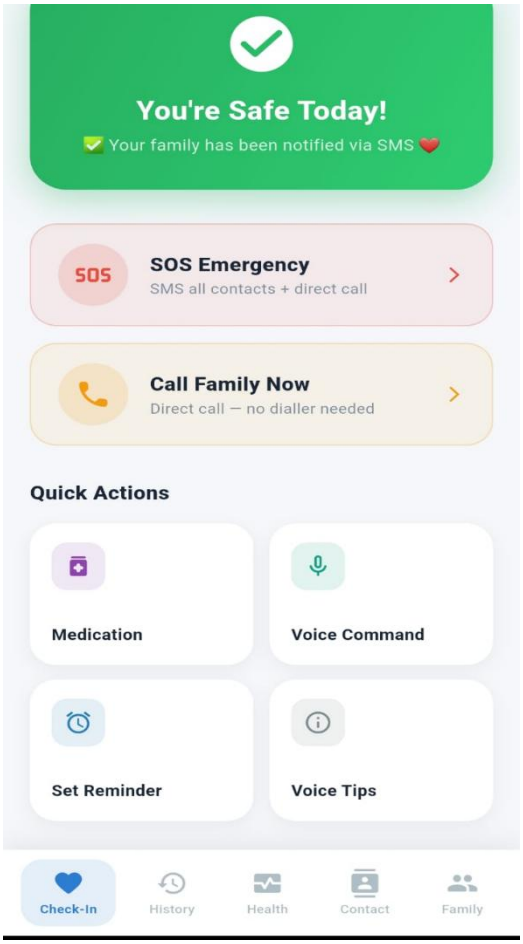

Figure 7.3 SeniorCare Dashboard

In fig 7.3 the lower section of the dashboard focuses on providing quick and easy access to essential daily features through the **Quick Actions** panel. This section is designed to be simple and user-friendly, especially for senior citizens, allowing them to perform important tasks with minimal effort.

It includes options such as **Medication**, which helps users manage and keep track of their medicines, and **Voice Command**, which enables hands-free interaction with the application. The **Set Reminder** feature allows users to schedule alerts for medications or daily activities, ensuring they do not miss important tasks. Additionally, the **Voice Tips** option provides helpful instructions and guidance through audio support, improving accessibility.

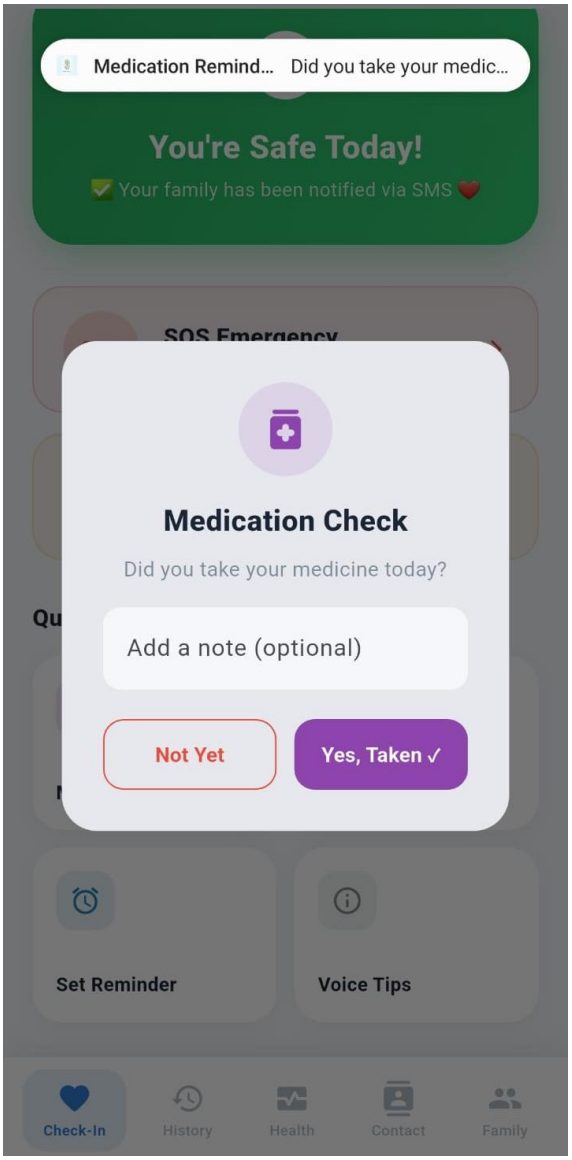

Figure 7.4 Medication Check Pop-up

In fig 7.4 The medication check pop-up is designed to ensure that users follow their prescribed medication schedule. It appears as a reminder asking the user whether they have taken their medicine for the day. The interface provides a simple input field to add an optional note, allowing users to record additional information such as dosage or timing.

Two clear options are provided: “Not Yet” and “Yes, Taken”, making it easy for elderly users to respond quickly. This feature helps in maintaining a proper medication log and ensures adherence to medical routines, which is crucial for health management.

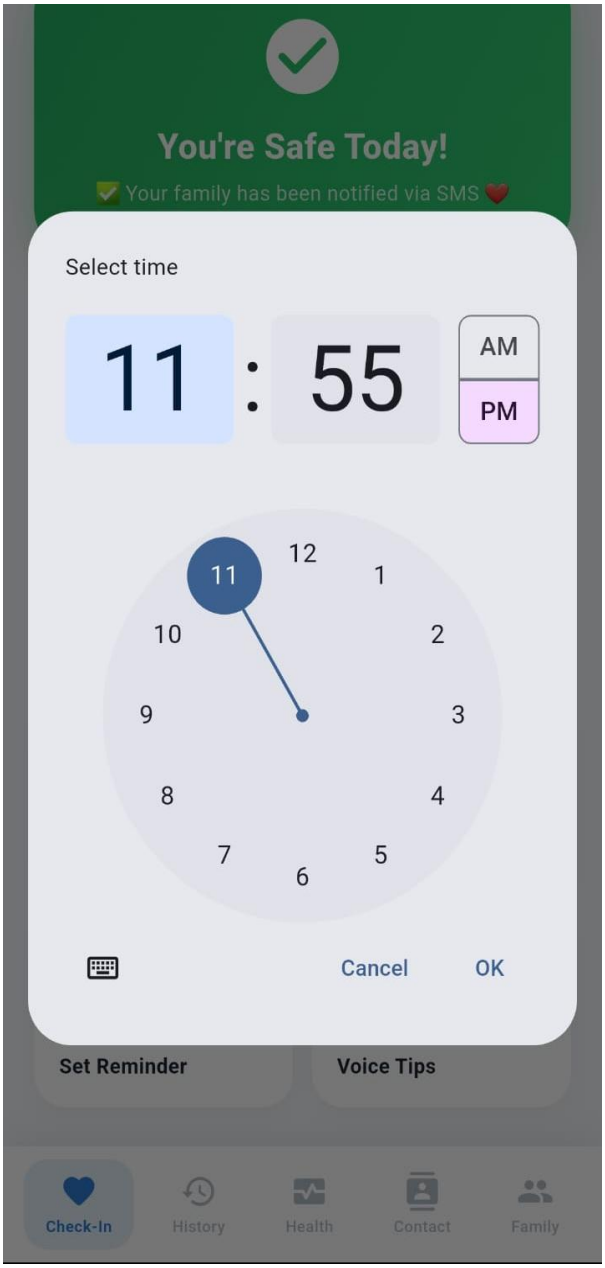

Figure 7.5 Set Reminder (Time Selection Interface)

In fig 7.5 the screen allows users to set reminders using an intuitive time selection interface. It includes both digital (hour and minute) and analog clock-style selection, making it easy for users with different preferences to choose a time.

The AM/PM toggle ensures correct scheduling, and users can confirm or cancel the selected time. This feature is essential for setting reminders for medications, appointments, or daily activities, promoting discipline and timely task completion.

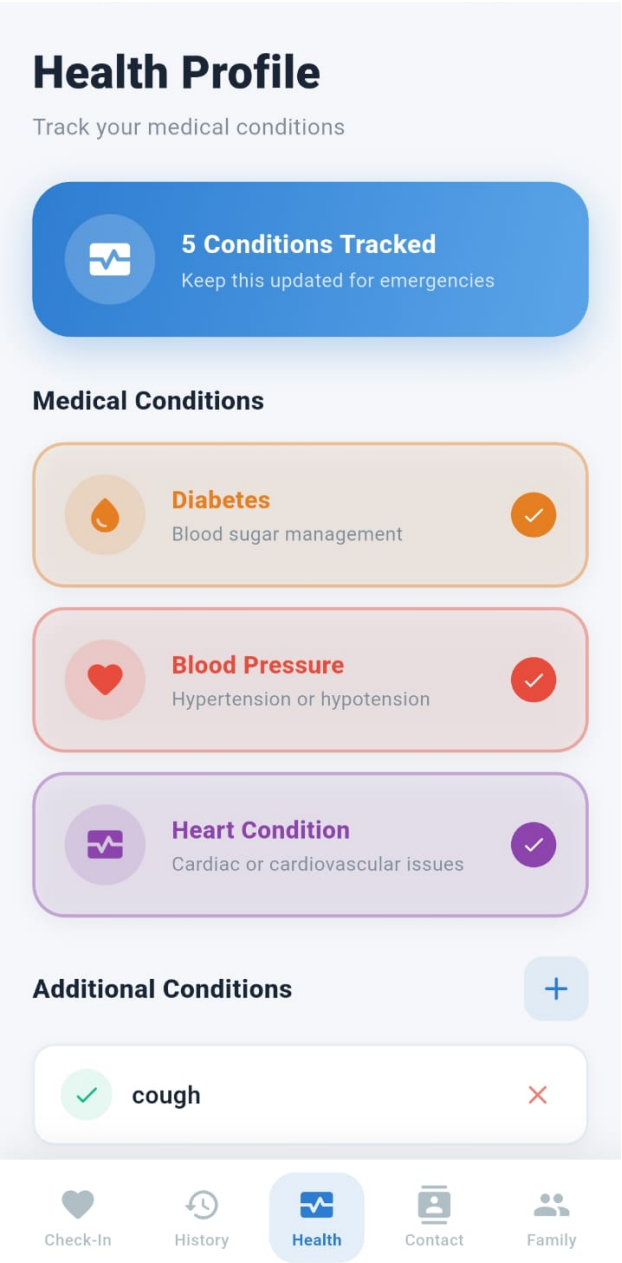

Figure 7.6 Health Profile Section

In fig 7.6 the health profile section provides a comprehensive overview of the user's medical conditions. It displays the number of conditions being tracked and emphasizes the importance of keeping this information updated for emergency situations.

Key conditions such as Diabetes, Blood Pressure, and Heart Condition are clearly listed with visual indicators. There is also an option to add additional conditions. This centralized health record helps caregivers and emergency responders quickly understand the user's medical background.

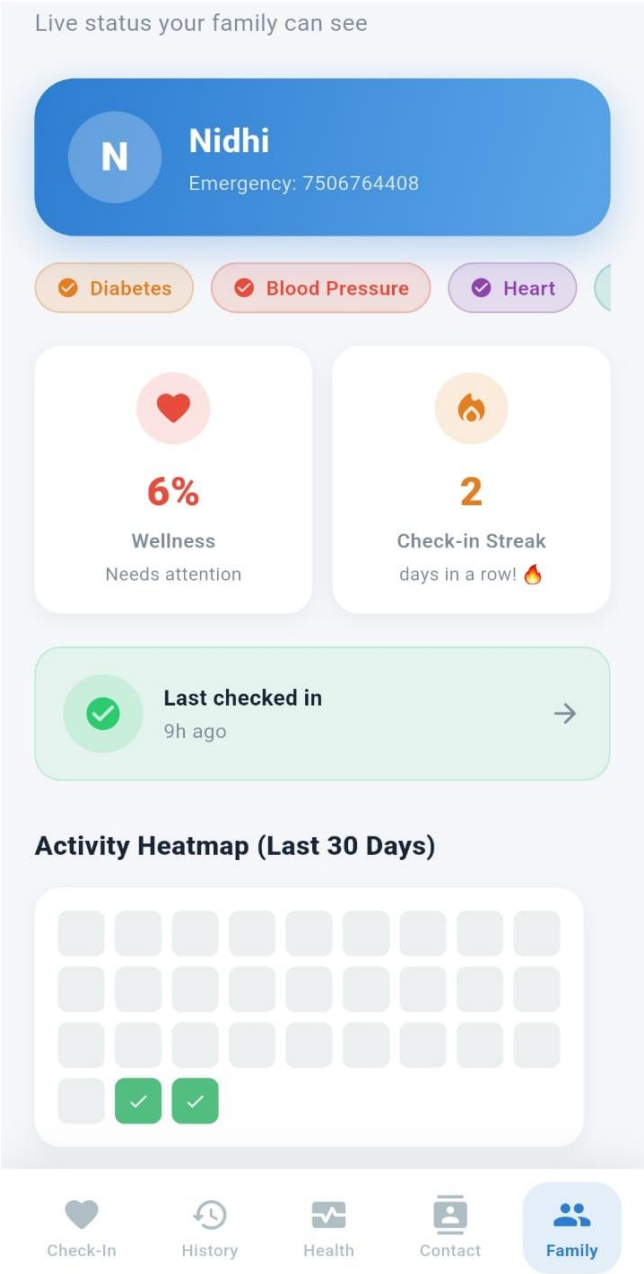

Figure 7.7 Emergency Contacts Section

In fig 7.7 the section manages the list of people to be contacted during emergencies. It clearly indicates that the primary contact receives direct SOS calls, while all contacts receive alerts.

Each contact entry includes details such as name, phone number, and relationship (e.g., family or friend). Options to call, edit, or delete contacts are provided. Additionally, users can add new emergency contacts easily. This ensures quick communication and immediate support during critical situations.

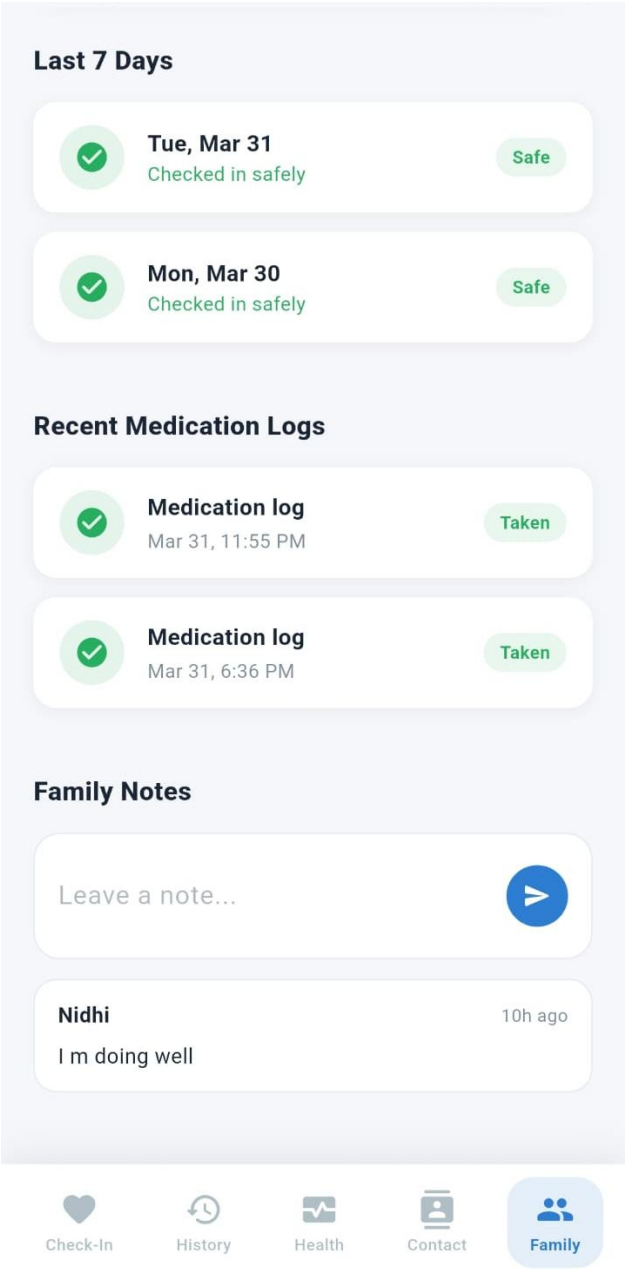

Figure 7.8 Family Dashboard / Live Status

In fig 7.8 the family dashboard allows family members to monitor the user's real-time status. It displays essential details such as the user's name, emergency contact, and current health indicators.

Metrics like wellness percentage, check-in streak, and last check-in time provide insights into the user's daily activity and condition. An activity heatmap visually represents user engagement over time. This feature strengthens connectivity and keeps family members informed about the user's well-being.

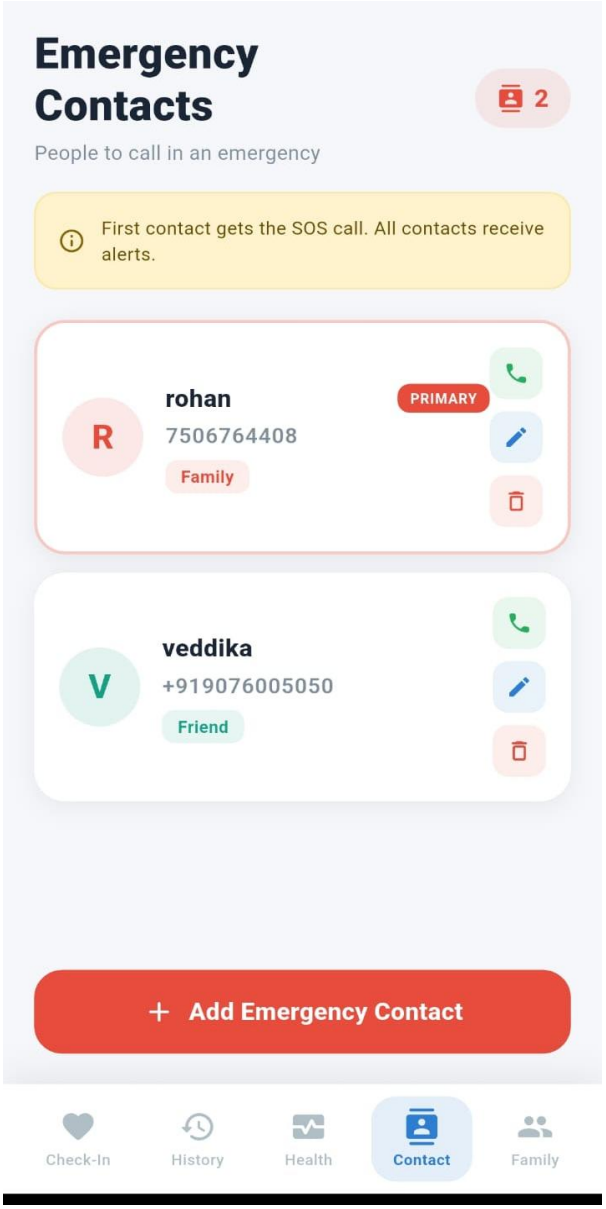

Figure 7.9 History and Logs Section

In fig 7.9 the section maintains a record of the user's recent activities, including check-ins and medication logs. It shows daily status updates such as "Checked in safely" along with timestamps, ensuring transparency and tracking.

Medication logs indicate whether medicines were taken, helping maintain a consistent health routine. Additionally, the Family Notes feature allows users and family members to communicate through short messages. This section acts as a historical record, aiding in monitoring health patterns and improving care management.

Chapter 8
Conclusion

The **Senior Connect system** offers a comprehensive and reliable solution for elderly care by combining safety features, real-time communication, and health monitoring within a single, user-friendly platform. It effectively addresses the challenges faced by senior citizens living independently by providing functionalities such as emergency alerts, daily check-ins, medication reminders, and instant communication with family members. These features significantly reduce risks by ensuring that help can be accessed quickly during emergencies, while also keeping caregivers continuously informed about the well-being of the user.

The system is designed with scalability in mind, allowing it to adapt and expand as technology evolves. It has the potential to be further enhanced through the integration of advanced technologies such as Artificial Intelligence (AI) and the Internet of Things (IoT). For instance, AI can be used for predictive health analysis to detect potential risks before they become critical, while IoT-enabled wearable devices can provide real-time health data such as heart rate, activity levels, and location tracking. Additionally, voice assistant integration can improve accessibility, making the system even more convenient for elderly users who may not be comfortable with manual interaction.

Overall, this project demonstrates how modern technological solutions can play a crucial role in improving elderly care. By ensuring safety, enabling independence, and strengthening the connection between seniors and their caregivers, the system enhances the overall quality of life for senior citizens. It highlights the importance and impact of digital innovation in creating a safer, more supportive environment for the aging population.

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